

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution Affiliated to Anna University)
Machine Learning Algorithms Laboratory

Degree & Branch	M.Tech (Integrated) CSE
Semester	V
Subject Code	ICS1512
Experiment	2
Title	Email Spam/Ham Classification using Naïve Bayes and KNN

Objective

- Build spam classifiers using Naïve Bayes and KNN.
- Compare Gaussian, Multinomial and Bernoulli Naïve Bayes.
- Study the effect of varying k .
- Compare KDTree and BallTree.
- Optimize KNN using GridSearchCV and RandomizedSearchCV.
- Evaluate computational complexity and execution time.
- Evaluate using K-Fold Cross Validation.

Theory

Naïve Bayes is a probabilistic classifier based on Bayes' theorem with conditional independence assumptions.

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

KNN is a lazy learning algorithm that classifies a sample based on the majority class among its nearest neighbors.

Dataset

Spambase Dataset from Kaggle: <https://www.kaggle.com/datasets/somesh24/spambase>

Software Requirements

Python, NumPy, Pandas, Matplotlib, Scikit-learn, Seaborn (optional).

Implementation Steps

1. Load dataset.
2. Handle missing values.
3. Normalize features.
4. Perform exploratory data analysis.
5. Split data into train and test sets.
6. Train:
 - Gaussian NB
 - Multinomial NB
 - Bernoulli NB
 - KNN
7. Tune KNN using GridSearchCV.
8. Tune KNN using RandomizedSearchCV.
9. Compare KDTree and BallTree.
10. Perform 5-fold Cross Validation.
11. Measure training and prediction time.
12. Draw evaluation plots.

Performance Metrics

Accuracy, Precision, Recall, F1-score, ROC-AUC, Confusion Matrix.

Required Tables

Naïve Bayes Comparison

Metric	Gaussian	Multinomial	Bernoulli
Accuracy			
Precision			
Recall			
F1			
ROC-AUC			

KNN Comparison

k	Accuracy	Precision	Recall	F1
1				
3				
5				
7				
9				
11				

Grid Search vs Randomized Search

Parameter	GridSearchCV	RandomizedSearchCV
Best k		
Metric		
Weights		
Algorithm		
CV Accuracy		
Execution Time		

KDTree vs BallTree

Metric	KDTree	BallTree
Accuracy		
Training Time		
Prediction Time		

Cross Validation

Fold	Naïve Bayes	Best KNN
1		
2		
3		
4		
5		
Average		

Theoretical Time Complexity

Algorithm	Training	Prediction
Naïve Bayes	$O(nd)$	$O(d)$
KNN (Brute)	$O(1)$	$O(nd)$
KDTree	$O(n \log n)$	$O(\log n)_{avg}$
BallTree	$O(n \log n)$	$O(\log n)_{avg}$

Experimental Time Analysis

Algorithm	Training(s)	Prediction(s)
Gaussian NB		
Multinomial NB		
Bernoulli NB		
Best KNN		

Required Plots

1. Class distribution.
2. Correlation heatmap.
3. Histograms of important features.
4. Boxplots.
5. Confusion matrix for each model.
6. ROC curves.
7. Precision-Recall curves.
8. Accuracy vs k .
9. Cross-validation accuracy.
10. Training time comparison.
11. Prediction time comparison.
12. Classifier comparison bar chart.
13. GridSearchCV heatmap.
14. RandomizedSearchCV score distribution.

Analysis Questions

1. Which Naïve Bayes variant performed best?
2. What is the optimal value of k ?
3. Compare GridSearchCV and RandomizedSearchCV.
4. Compare KDTree and BallTree.
5. Compare theoretical and practical complexity.
6. Which classifier is preferred for large datasets?

Additional Tasks

- Evaluate different train-test splits.
- Compare Euclidean and Manhattan distances.
- Evaluate weighted KNN.
- Discuss overfitting for small k and underfitting for large k .

Report Checklist

- Aim and Objective
- Theory
- Dataset
- EDA
- Code
- Hyperparameter tuning
- Comparison tables
- Complexity analysis
- Plots
- Results
- Conclusion

Viva Questions

1. State Bayes theorem.
2. Difference between Gaussian and Multinomial NB.
3. Why normalize features for KNN?
4. Difference between KDTree and BallTree.
5. Why use cross-validation?
6. Difference between GridSearchCV and RandomizedSearchCV.
7. Explain ROC-AUC.
8. Explain Precision and Recall.
9. Why does KNN have negligible training time?
10. Why does prediction become slower for KNN?

References

- Scikit-learn Documentation.
- Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow.
- Pattern Recognition and Machine Learning.